Compte rendu du TP : API e-commerce NestJS + MySQL + Docker + Swagger

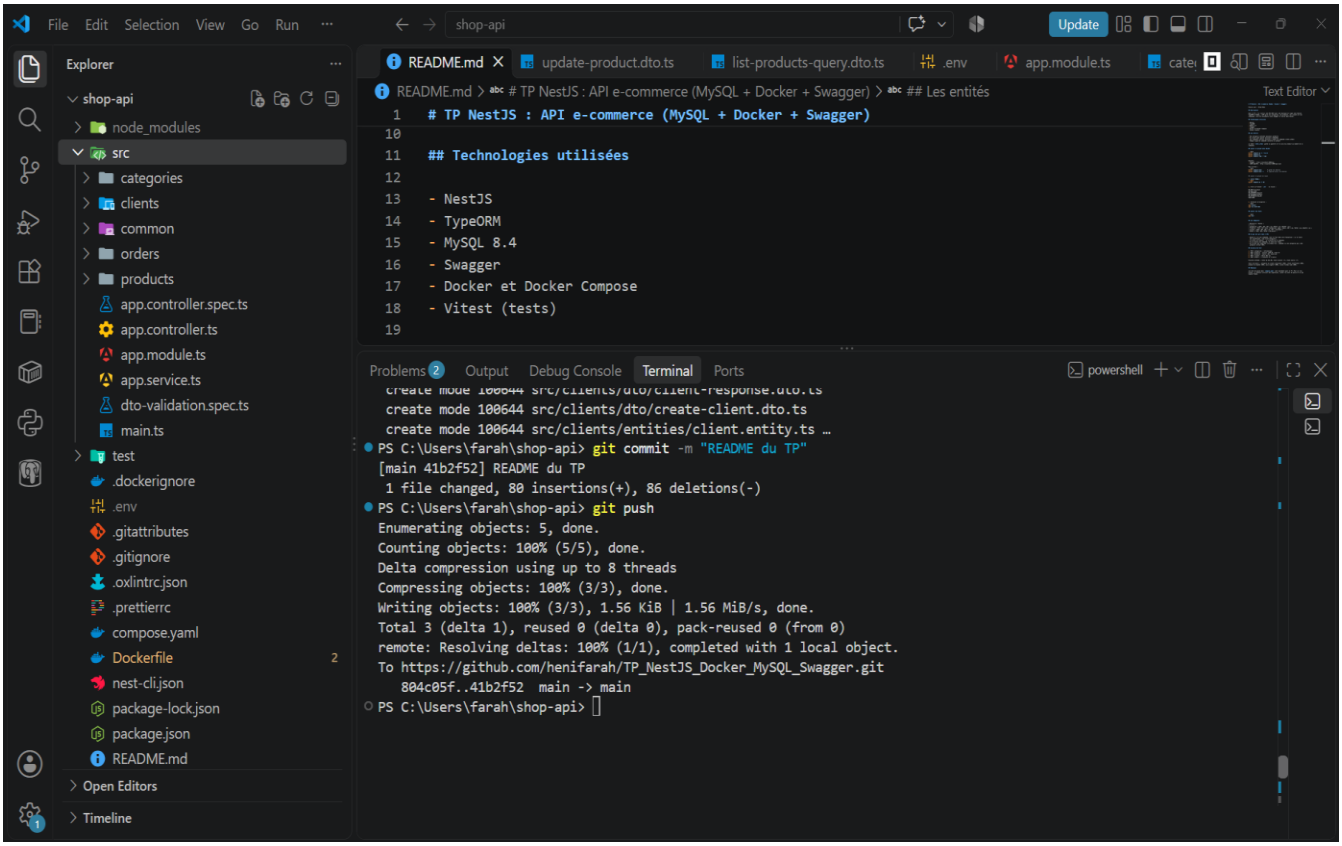
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Dépôt : https://github.com/henifarah/TP_NestJS_Docker_MySQL_Swagger

1. Installation et structure :

J'ai installé NestJS, TypeORM, MySQL, Swagger et class-validator, puis créé 4 modules (categories, products, clients, orders). Chacun a un controller, un service, un dossier `dto` et un dossier `entities`.

Une erreur rencontrée : `type: module` dans `package.json` faisait planter le démarrage. Je l'ai supprimé et configuré `tsconfig.json` en CommonJS.



2. Entités et relations :

Relation	Clé étrangère
Category 1-N Product	products.categoryId
Client 1-N CustomerOrder	customer_orders.clientId
CustomerOrder 1-N OrderItem	order_items.orderId

Relation	Clé étrangère
Product 1-N OrderItem	order_items.productId

La clé étrangère est toujours du côté ManyToOne. La table `order_items` stocke la quantité et le prix figé (`unitPrice`) : c'est une relation N-N enrichie. `RESTRICT` protège l'historique, `CASCADE` supprime les lignes avec leur commande.

```
PS C:\Users\farah\shop-api> docker compose exec db mysql -u shopuser -pshoppass shop_db
mysql> SHOW TABLES;
+------------------+
| Tables_in_shop_db |
+------------------+
| categories       |
| clients          |
| customer_orders  |
| order_items      |
| products         |
+------------------+
5 rows in set (0.00 sec)

mysql> DESCRIBE products;
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| id         | int           | NO   | PRI | NULL    | auto_increment |
| name       | varchar(255)  | NO   |     | NULL    |                |
| price      | decimal(10,2) | NO   |     | NULL    |                |
| stock      | int           | NO   |     | 0       |                |
| categoryId | int           | NO   | MUL | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
5 rows in set (0.01 sec)

mysql> DESCRIBE order_items;
+-----------+---------------+------+-----+---------+----------------+
| Field     | Type          | Null | Key | Default | Extra          |
+-----------+---------------+------+-----+---------+----------------+
| id        | int           | NO   | PRI | NULL    | auto_increment |
| quantity  | int           | NO   |     | NULL    |                |
| unitPrice | decimal(10,2) | NO   |     | NULL    |                |
| orderId   | int           | NO   | MUL | NULL    |                |
| productId | int           | NO   | MUL | NULL    |                |
+-----------+---------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

3.

DTO, validation et erreurs :

Chaque création a son DTO avec class-validator (champ obligatoire, prix ≥ 0, email valide, liste d'items non vide). Le ValidationPipe global refuse les champs inconnus.

Erreur	Code
Données invalides, stock insuffisant, produit en double	400
Catégorie, client, produit ou commande introuvable	404
Suppression d'un élément encore utilisé	409

4. Services et endpoints :

Ressource	Routes
Categories	POST, GET, GET /:id, PATCH /:id, DELETE /:id
Products	POST, GET (?categoryId, page, limit), GET /:id, PATCH /:id, DELETE /:id
Clients	POST, GET, GET /:id, GET /:id/orders
Orders	POST, GET, PATCH /:id/cancel

5. Commande transactionnelle et stock :

La commande est créée dans une transaction : si une règle échoue, rien n'est enregistré.

- Le stock baisse avec un `UPDATE` conditionnel (`stock >= quantité`), donc il ne peut pas devenir négatif, même avec deux commandes simultanées.
- Le total est calculé en centimes et le prix est copié dans `unitPrice`.
- L'annulation remet le stock, dans une transaction aussi.

Résultat	Attendu	Obtenu
Total de la commande	321.00	à compléter
Stock Clavier / Souris	6 / 9	à compléter

Curl

```
curl -X 'POST' \
  'http://localhost:3000/orders' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "clientId": 1,
  "items": [
    {
      "productId": 1,
      "quantity": 2
    }
  ]
}'
```

Request URL

```
http://localhost:3000/orders
```

Server response

Code	Details
201	Response body

```
{
  "id": 2,
  "status": "CREATED",
  "total": "201.00",
  "createdAt": "2026-10-26T12:05:05.535Z",
  "client": {
    "id": 1,
    "fullName": "Amira Ben Ali",
    "email": "amira@example.com"
  },
  "clientId": 1,
  "items": [
    {
      "id": 3,
      "quantity": 2,
      "unitPrice": "100.50",
      "orderId": 2,
      "product": {
        "id": 1,
        "name": "Clavier",
        "price": "100.50",
        "stock": 4,
        "categoryId": 1
      },
      "productId": 1
    }
  ]
}
```

Response headers

```
mysql> SHOW TABLES;
+-------------------+
| Tables_in_shop_db |
+-------------------+
| categories        |
| clients           |
| customer_orders   |
| order_items       |
| products          |
+-------------------+
5 rows in set (0.01 sec)

mysql> DESCRIBE products;
+------------+---------------+------+-----+---------+----------------+
| Field      | Type          | Null | Key | Default | Extra          |
+------------+---------------+------+-----+---------+----------------+
| id         | int           | NO   | PRI | NULL    | auto_increment |
| name       | varchar(255)  | NO   |     | NULL    |                |
| price      | decimal(10,2) | NO   |     | NULL    |                |
| stock      | int           | NO   |     | 0       |                |
| categoryId | int           | NO   | MUL | NULL    |                |
+------------+---------------+------+-----+---------+----------------+
5 rows in set (0.01 sec)

mysql> DESCRIBE order_items;
+-----------+---------------+------+-----+---------+----------------+
| Field     | Type          | Null | Key | Default | Extra          |
+-----------+---------------+------+-----+---------+----------------+
| id        | int           | NO   | PRI | NULL    | auto_increment |
| quantity  | int           | NO   |     | NULL    |                |
| unitPrice | decimal(10,2) | NO   |     | NULL    |                |
| orderId   | int           | NO   | MUL | NULL    |                |

mysql> DESCRIBE order_items;
+-----------+---------------+------+-----+---------+----------------+
| Field     | Type          | Null | Key | Default | Extra          |
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| id        | int           | NO   | PRI | NULL    | auto_increment |
| quantity  | int           | NO   |     | NULL    |                |
| unitPrice | decimal(10,2) | NO   |     | NULL    |                |
| orderId   | int           | NO   | MUL | NULL    |                |
| productId | int           | NO   | MUL | NULL    |                |
+-----------+---------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM categories;
+----+--------------+
| id | name         |
+----+--------------+
|  3 | cloud        |
|  1 | Informatique |
+----+--------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM products;
+----+--------------------+--------+-------+------------+
| id | name               | price  | stock | categoryId |
+----+--------------------+--------+-------+------------+
|  1 | Clavier            | 100.50 |     2 |          1 |
|  2 | Souris             |  40.00 |     9 |          1 |
|  3 | Clavier mecanique  | 150.00 |    12 |          1 |
|  4 | Clavier mecanique  | 150.00 |    12 |          1 |
|  5 | Clavier mecanique  | 150.00 |    12 |          1 |
|  6 | Clavier mecanique  | 150.00 |    12 |          1 |
+----+--------------------+--------+-------+------------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM clients;
+----+---------------+-------------------+
| id | fullName      | email             |
+----+---------------+-------------------+
|  1 | Amira Ben Ali | amira@example.com |
+----+---------------+-------------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM customer_orders;
+----+---------+--------+----------------------------+----------+
| id | status  | total  | createdAt                  | clientId |
+----+---------+--------+----------------------------+----------+
|  1 | CREATED | 321.00 | 2026-10-26 09:43:48.420503 |        1 |
|  2 | CREATED | 201.00 | 2026-10-26 12:05:05.535497 |        1 |
|  3 | CREATED | 201.00 | 2026-11-10 10:09:30.495325 |        1 |
+----+---------+--------+----------------------------+----------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM order_items;
+----+----------+-----------+---------+-----------+
| id | quantity | unitPrice | orderId | productId |
+----+----------+-----------+---------+-----------+
|  1 |        2 |    100.50 |       1 |         1 |
|  2 |        3 |     40.00 |       1 |         2 |
|  3 |        2 |    100.50 |       2 |         1 |
|  4 |        2 |    100.50 |       3 |         1 |
+----+----------+-----------+---------+-----------+
4 rows in set (0.00 sec)

mysql>
```